

# Management of Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer and Brain Metastases: ASCO Guideline Update Q and A

Naren Ramakrishna, MD, PhD<sup>1</sup>; Carey K. Anders, MD<sup>2</sup>; and Sarah Temin, MSPH<sup>3</sup>

ASCO recently published an updated guideline for the management of advanced human epidermal growth factor receptor 2 (HER2)–positive breast cancer and brain metastases,<sup>1</sup> along with a companion guideline on systemic therapy for patients with HER2-positive advanced breast cancer.<sup>2</sup> Recent data on local therapy and systemic therapy advances for patients with HER2-positive advanced breast cancer and brain metastases drove updates to the guideline recommendations, previously updated in 2018.<sup>3</sup> In this article, guideline authors address the nuances of selecting the appropriate treatment and surveillance regimens for these patients (Fig 1).

## QUESTION: HOW DO YOU SELECT PATIENTS WITH HER2-POSITIVE BREAST CANCER AND BRAIN METASTASES FOR WHOM UPFRONT SYSTEMIC THERAPY IS APPROPRIATE AS AN ALTERNATIVE TO LOCAL THERAPY?

Naren Ramakrishna, MD, PhD: Recent randomized phase III clinical trial data (HER2CLIMB) have significantly changed the upfront local therapy paradigm for patients with HER2-positive breast cancer and brain metastases, which are asymptomatic.<sup>4,5</sup> In the brain metastasis subgroup, treatment with tucatinib + capecitabine + trastuzumab resulted in an intracranial objective response rate of 47.3% versus 20% in the control arm with a median intracranial progression-free survival of 9.9 months versus 4.2 months.<sup>4</sup> Given these data, it is reasonable, following multidisciplinary review with neurosurgery and radiation oncology and discussion with the patient, to offer the option of upfront systemic therapy and deferral of local therapy such as stereotactic radiosurgery (SRS) or whole-brain radiotherapy (WBRT) for patients with favorable prognosis with asymptomatic brain metastases. Local therapy with surgery, SRS, or WBRT has historically been the mainstay of upfront therapy, and the choice to pursue systemic therapy and defer local therapy must be carefully weighed, especially for tumors in critical locations, or larger tumors, where a small amount of additional growth may result in potentially irreversible neurological symptoms. In these

higher-risk settings, even in asymptomatic patients, the use of local therapy is preferred.

## QUESTION: FOR PATIENTS UNSUITABLE FOR SURGERY OR SYSTEMIC THERAPY IN WHOM LOCAL THERAPY WITH RADIATION IS RECOMMENDED, HOW DO YOU SELECT THE APPROPRIATE MODALITY BETWEEN SRS AND WBRT?

Naren Ramakrishna, MD, PhD: In recent years, there has been an increasing shift from WBRT ± SRS toward SRS alone for patients with discrete (as opposed to diffuse or leptomeningeal) brain metastatic disease. WBRT was traditionally recommended for patients with multiple brain metastases, but an increasing body of data support the use of SRS alone for better preservation of function.<sup>6,7</sup> Furthermore, there are emerging data to support the use of SRS for larger numbers of brain metastases compared with the historical normal limit of ≤ 4.<sup>8</sup> These data have resulted in increased usage of SRS as a primary local therapy with WBRT deferred for cases where SRS is not feasible such as diffuse or leptomeningeal presentations or multiple large metastases.

Although the use of WBRT has declined, the risk-benefit ratio of WBRT has been improved in recent years through the use of neuroprotectants such as memantine, which resulted in a 22% relative risk reduction for cognitive decline in a randomized double-blinded placebo-controlled trial when administered concurrently and for 6 months following WBRT compared with WBRT alone.<sup>9</sup> The use of advanced radiation dosimetric techniques for WBRT resulting in reduced dose to the hippocampal region together with memantine (WBRT + memantine + hippocampal avoidance) was evaluated in a phase III randomized trial versus conventional WBRT + memantine and demonstrated a 26% relative risk reduction for cognitive failure with reductions also in fatigue and improvement in activities of daily living.<sup>10</sup> There have been no head-to-head studies of the improved WBRT techniques versus SRS alone for multiple metastases, but for those patients recommended for WBRT, these techniques offer a means to decrease potential toxicity.

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on June 28, 2022 and published at [ascopubs.org/journal/op](https://ascopubs.org/journal/op) on August 10, 2022; DOI <https://doi.org/10.1200/OP.22.00364>

| Recommendation                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type  | Evidence Quality | Strength |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------|----------|
| <b>2.3.</b> If a patient has a favorable prognosis and a single brain metastasis < 2 cm without symptomatic mass effect and who has an option to proceed with HER2-directed therapy with known CNS activity, then clinicians and patients may discuss options including SRS or deferring local therapy with a MDT.                                                                                                                                                                | IC    | L                | M        |
| <b>3.0.</b> If a patient has a favorable prognosis and presents with multiple, but limited, metastases (defined as two to four lesions), treatment options depend on the size, resectability, and mass effect of the lesions.                                                                                                                                                                                                                                                     | NA    | NA               | NA       |
| <b>3.1.</b> In a patient who presents with limited metastases [defined as two to four lesions] suitable for SRS, clinicians may discuss SRS without WB-M + HA.                                                                                                                                                                                                                                                                                                                    | FC    | I                | W        |
| <b>3.2.</b> In a patient with symptomatic lesions that are unresectable and unsuitable for SRS or HSRT, clinicians may recommend WBRT plus memantine and, if feasible, hippocampal avoidance and may discuss SRS after WB-M + HA.                                                                                                                                                                                                                                                 | FC/IC | L                | W        |
| <b>3.3.</b> For patients with limited metastases < 2 cm and not associated with symptomatic mass effect, and who have an option to proceed with HER2-directed therapy with known CNS activity, then clinicians and patients may discuss deferring local therapy with a MDT.                                                                                                                                                                                                       | IC    | L                | M        |
| <b>3.4.</b> In a patient who has a large (> 3 to 4 cm) lesion associated with symptomatic mass effect, clinicians may discuss surgical resection of the larger lesion, if the lesion is deemed resectable. The remaining lesions and resection bed may be treated with SRS, HSRT with or without WB-M + HA. Clinicians should also provide symptom management.                                                                                                                    | FC    | I                | W        |
| <b>4.2.1.</b> If a patient has a more favorable prognosis and presents with many diffuse and/or extensive brain metastases (≥ five metastases) without leptomeningeal disease, clinicians may recommend SRS or WB-M + HA. For patients with metastases < 2 cm and not associated with symptomatic mass effect, and who have an option to proceed with HER2-directed therapy with known CNS activity, then clinicians and patients may discuss deferring local therapy with a MDT. | FC/IC | L                | M        |
| <b>7.1.</b> The combination of tucatinib, and capecitabine and trastuzumab may be offered to patients with HER2 positive metastatic breast cancer who have brain metastases without symptomatic mass effect and whose disease has progressed on at least one previous HER2-directed therapy for metastatic disease. If these agents are used, local therapy may be delayed until there is evidence of intracranial progression.                                                   | EB    | L                | W        |
| <b>8.1.</b> For a patient who receives a standard surgical- or radiotherapy-based approach to treat brain metastases and is receiving anti-HER2-based therapy and whose systemic disease is not progressive at the time of brain metastasis diagnosis, clinicians should not switch systemic therapy.                                                                                                                                                                             | FC    | L                | M        |

**FIG 1.** Summary of key updated guideline recommendations. EB, evidence-based; FC, formal consensus; HA, hippocampal avoidance; HER2, human epidermal growth factor receptor 2; HSRT, hypofractionated stereotactic radiotherapy; I, intermediate; IC, informal consensus; L, low; M, moderate; MDT, multidisciplinary team; NA, not applicable; SRS, stereotactic radiosurgery; W, weak; WB-M, whole-brain radiation therapy and memantine; WBRT, whole brain radiotherapy.

**QUESTION: HOW DOES ONE CHOOSE BETWEEN TRASTUZUMAB DERUXTECAN VERSUS THE HER2CLIMB REGIMEN (TUCATINIB + CAPECITABINE + TRASTUZUMAB) IN THE SETTING OF HER2-POSITIVE BREAST CANCER AND BRAIN METASTASES AFTER ONE PRIOR HER2-DIRECTED THERAPY?**

Carey Anders, MD: There are many factors that play into this decision. First is the status of the brain metastases as “active” (either previously untreated with local therapy or progressive after local therapy) or “stable” following local therapy. The most robust data for “active” brain metastases are with the HER2CLIMB regimen based on randomized, phase III, placebo-controlled data illustrating the superiority of tucatinib + capecitabine + trastuzumab in this setting. There are emerging data for trastuzumab deruxtecan in this setting, but largely based on observational data and smaller, single-arm studies. Thus, in the setting of “active” brain metastases and in the absence of contraindications, the guidelines recommend the HER2CLIMB regimen for the treatment of “active” brain metastases. In the setting of “stable” brain metastases, there are robust data for both trastuzumab deruxtecan and the HER2CLIMB regimen; a choice between these two regimens in this setting is based

on patient characteristics, tolerance, preference, financial toxicity, and capacity for amenability.

**QUESTION: IN THE ADVENT OF SO MANY NEW THERAPIES FOR HER2-POSITIVE BRAIN METASTASES, WHY IS SCREENING MAGNETIC RESONANCE IMAGING IN THE ABSENCE OF SYMPTOMS NOT RECOMMENDED ROUTINELY?**

Carey Anders, MD: Although the landscape of brain-permeable, HER2-targeted agents has expanded, we do not yet have robust data illustrating that routine brain magnetic resonance imaging (MRI) in the absence of symptoms improves patient survival in a meaningful way. I think we all believe that earlier detection will ultimately lead to improvements in outcome; however, we are in a data-free zone and may be subject to lead-time bias. There are ongoing, prospective studies evaluating the impact of screening MRIs on patient outcomes in this setting. Until we have these data, the panel updated the guidelines to indicate there are insufficient data to recommend for or against routine brain MRI screening in the absence of symptoms. One should have a very low threshold for ordering a brain MRI if any neurologic symptoms are reported in this setting, which is what I do in my own practice.

## AFFILIATIONS

<sup>1</sup>Orlando Health Cancer Institute, Orlando, FL

<sup>2</sup>Duke University, Durham, NC

<sup>3</sup>American Society of Clinical Oncology, Alexandria, VA

## CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Rd, Suite 800, Alexandria, VA 22314; e-mail: [guidelines@asco.org](mailto:guidelines@asco.org).

## DISCLAIMER

This Q&A is derived from recommendations in Management of Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer and Brain Metastases: ASCO Guideline Update. This document is based on an ASCO Guideline and is not intended to substitute for the independent professional judgment of the treating physician. Practice guidelines do not account for individual variation among patients. This Q&A does not purport to suggest any particular course of medical treatment. Use of the guideline and this Q&A is voluntary. Please refer to the complete guideline to understand the full recommendations.

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/OP.22.00364>.

## REFERENCES

- Ramakrishna N, Anders CK, Lin NU: Recommendations on disease management for patients with advanced human epidermal growth factor receptor 2–positive breast cancer and brain metastases: ASCO guideline update. *J Clin Oncol* 40:2636-2655, 2022
- Giordano S, Franzoi MAB, Temin S, et al: Systemic therapy for patients with advanced human epidermal growth factor receptor 2–positive breast cancer: ASCO guideline update. *J Clin Oncol* 40:2612-2635, 2022
- Ramakrishna N, Temin S, Chandralapaty S, et al: Recommendations on disease management for patients with advanced human epidermal growth factor receptor 2–positive breast cancer and brain metastases: ASCO clinical practice guideline update. *J Clin Oncol* 36:2804-2807, 2018
- Lin NU, Borges V, Anders C, et al: Intracranial efficacy and survival with tucatinib plus trastuzumab and capecitabine for previously treated HER2-positive breast cancer with brain metastases in the HER2CLIMB trial. *J Clin Oncol* 38:2610-2619, 2020
- Murthy RK, Loi S, Okines A, et al: Tucatinib, trastuzumab, and capecitabine for HER2-positive metastatic breast cancer. *N Engl J Med* 382:597-609, 2020
- Brown PD, Jaeckle K, Ballman KV, et al: Effect of radiosurgery alone vs radiosurgery with whole brain radiation therapy on cognitive function in patients with 1 to 3 brain metastases: A randomized clinical trial. *JAMA* 316:401-409, 2016
- Chang EL, Wefel JS, Hess KR, et al: Neurocognition in patients with brain metastases treated with radiosurgery or radiosurgery plus whole-brain irradiation: A randomised controlled trial. *Lancet Oncol* 10:1037-1044, 2009
- Yamamoto M, Serizawa T, Shuto T, et al: Stereotactic radiosurgery for patients with multiple brain metastases (JLGK0901): A multi-institutional prospective observational study. *Lancet Oncol* 15:387-395, 2014
- Brown PD, Pugh S, Laack NN, et al: Memantine for the prevention of cognitive dysfunction in patients receiving whole-brain radiotherapy: A randomized, double-blind, placebo-controlled trial. *Neuro Oncol* 15:1429-1437, 2013
- Brown PD, Gondi V, Pugh S, et al: Hippocampal avoidance during whole-brain radiotherapy plus memantine for patients with brain metastases: Phase III trial NRG oncology CC001. *J Clin Oncol* 38:1019-1029, 2020

## AUTHOR CONTRIBUTIONS

**Conception and design:** Naren Ramakrishna, Sarah Temin

**Administrative support:** Sarah Temin

**Collection and assembly of data:** Naren Ramakrishna, Sarah Temin

**Data analysis and interpretation:** Naren Ramakrishna, Carey K. Anders, Sarah Temin

**Manuscript writing:** All authors

**Final approval of manuscript:** All authors

**Accountable for all aspects of the work:** All authors

## ACKNOWLEDGMENT

Management of Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer and Brain Metastases: ASCO Guideline Update was developed and written by Naren Ramakrishna, MD, PhD; Carey K. Anders, MD; Nancy U. Lin, MD; Aki Morikawa, MD, PhD; Sarah Temin, MSPH; Sarat Chandralapaty, MD, PhD; Jennie R. Crews, MD; Nancy E. Davidson, MD; Maria Alice B. Franzoi, MD; Jeffrey J. Kirshner, MD; Ian E. Krop, MD, PhD; Debra A. Patt, MD, MPH, MBA; Jane Perlmutter, PhD; and Sharon H. Giordano, MD, MPH.



**AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST****Management of Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer and Brain Metastases: ASCO Guideline Update Q and A**

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to [www.asco.org/rwc](http://www.asco.org/rwc) or [ascopubs.org/op/authors/author-center](http://ascopubs.org/op/authors/author-center).

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](#)).

**Naren Ramakrishna**

**Stock and Other Ownership Interests:** Cytecare Cancer Center, Bangalore India

**Research Funding:** Decimal (Inst)

**Carey K. Anders**

**Honoraria:** Eisai, Genentech/Roche, Ipsen, Seattle Genetics, Puma Biotechnology, AstraZeneca, Elucida Oncology, Immunomedics, Athenex

**Consulting or Advisory Role:** Genentech/Roche, Puma Biotechnology, Eisai, Ipsen, Seattle Genetics, AstraZeneca, Elucida Oncology, Immunomedics

**Research Funding:** Puma Biotechnology (Inst), Lilly (Inst), Merck (Inst), Nektar (Inst), Tesaro (Inst), Seattle Genetics (Inst), G1 Therapeutics (Inst), Pfizer (Inst), ZION (Inst), Novartis Pharmaceuticals UK Ltd (Inst)

**Patents, Royalties, Other Intellectual Property:** UpToDate.com, Jones and Bartlett

**Travel, Accommodations, Expenses:** Eisai

No other potential conflicts of interest were reported.